

1. Statement of the Identification Problem

The central question of this paper is whether the public release of ChatGPT in October 2022 acted as a shock that reduced the share of entry-level job postings in occupations highly exposed to generative AI. A simple comparison of entry-level hiring trends across high- and low-exposure occupations is insufficient for causal identification.

AI-exposed occupations are structurally different from less-exposed ones. Occupations with high GPT-4 exposure scores (e.g., software developers, database administrators, mathematicians) are concentrated in the technology sector, which has historically maintained lower entry-level hiring shares, higher educational requirements, and distinct cyclical hiring patterns. Any naive before-after comparison conflates the effect of generative AI with pre-existing structural differences.

The identification strategy must therefore isolate differential changes in entry-level hiring *within* occupations around the ChatGPT launch, net of all time-invariant occupation characteristics and aggregate time-varying shocks.

2. Directed Acyclic Graph (DAG)

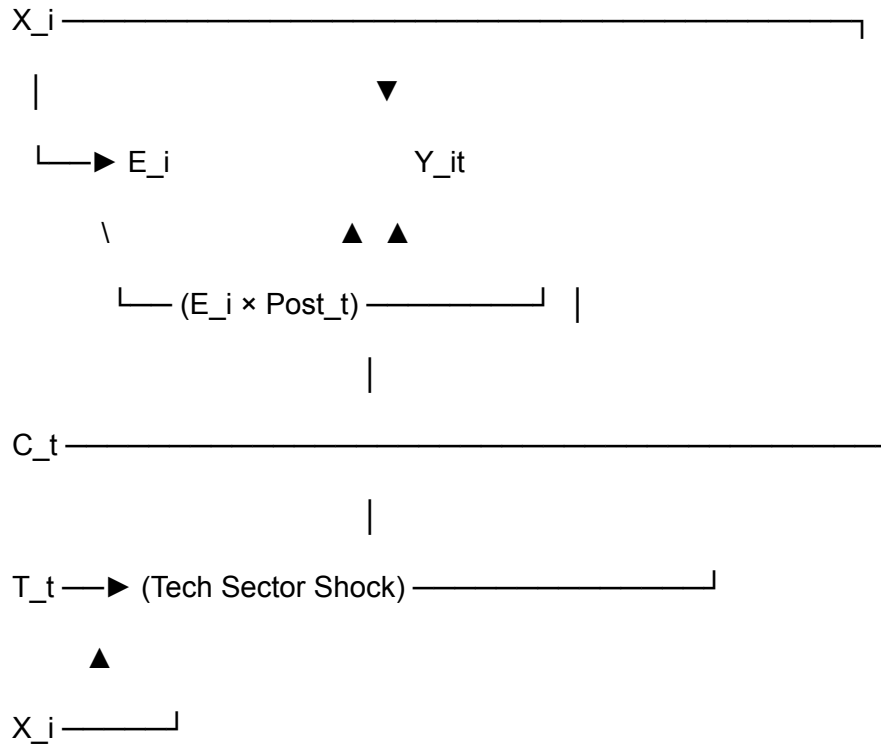
The DAG below represents the assumed causal structure of the problem. Nodes are:

- **E_i** — GPT-4 exposure score (occupation-level, time-invariant)
- **Post_t** — indicator for post-ChatGPT period (= 1 after October 2022)
- **Y_{it}** — entry-level share of job postings (occupation × month)
- **X_i** — time-invariant occupation characteristics (sector, skill intensity, wage level)
- **C_t** — aggregate economic conditions (macro shocks, labor market cycle)
- **T_t** — tech-sector-specific shock (2022–2023 layoffs)

Key paths and their interpretation:

- **X_i → E_i**: Occupation characteristics (sector, task content) determine AI exposure. This means exposure is not randomly assigned — it is correlated with everything else about an occupation.
- **X_i → Y_{it}**: The same characteristics that drive exposure also independently drive entry-level hiring patterns, creating confounding.
- **E_i × Post_t → Y_{it}**: The causal path of interest. The launch of ChatGPT (Post_t) activates the effect of AI exposure on entry-level hiring.
- **C_t → Y_{it}**: Aggregate shocks affect all occupations and must be absorbed by time fixed effects.
- **T_t → Y_{it}**: The tech-sector contraction (T_t) flows through sector identity (X_i) to affect high-exposure occupations disproportionately. This is the key threat to identification and is *not* blocked by standard two-way fixed effects.

The two-way fixed effects (TWFE) DiD design blocks the path through X_i by conditioning on occupation fixed effects (absorbing all time-invariant confounders) and the path through C_t by conditioning on time fixed effects. The residual threat is T_t , which is both time-varying and correlated with E_i .



3. Key Model Equations

3.1 Outcome Variable Construction

For each occupation i and month t , the seasonally adjusted entry-level share is:

$$Y_{it} = (\text{Entry-Level Postings}_{it} / \text{Total Postings}_{it}) - \text{seasonal component}$$

where the seasonal component is estimated by regressing the raw share on 11 month-of-year dummies and taking residuals. The resulting series is then indexed to October 2022 = 100 within each exposure group to allow cross-group comparison of growth rates.

3.2 Primary Difference-in-Differences Specification

The main estimating equation is a continuous-treatment DiD:

$$Y_{it} = \alpha + \beta * (\text{Post}_t \times E_i) + \gamma_i + \delta_t + \epsilon_{it}$$

where:

- Y_{it} is the indexed, seasonally adjusted entry-level share for occupation i in month t
- $Post_t = 1$ for months after October 2022, 0 otherwise
- E_i is the GPT-4 beta exposure score (continuous, 0–1), time-invariant
- γ_i are occupation fixed effects, absorbing all time-invariant occupation characteristics (sector, baseline hiring practices, wage level, task content)
- δ_t are month-year fixed effects, absorbing all aggregate shocks common across occupations
- ϵ_{it} is an idiosyncratic error term

beta is the coefficient of interest. A negative and significant beta would indicate that each additional unit of GPT-4 exposure is associated with a relative decline in entry-level hiring following ChatGPT's launch, compared to the counterfactual absent-ChatGPT trend.

Estimation is by weighted least squares (WLS), weighting each observation by the occupation's fixed 2021 posting share (w_{2021_pct}), to account for differences in market size across occupations. Standard errors are clustered by occupation to allow for within-occupation serial correlation over time.

3.3 Event-Study Specification

To assess pre-trends and trace the dynamic response to the ChatGPT shock, I estimate:

$$Y_{it} = \alpha + \sum_{k \neq -1} \beta_k * (1[t = k] \times E_i) + \gamma_i + \delta_t + \epsilon_{it}$$

where k indexes event time in months relative to October 2022 ($k = 0$), and $k = -1$ is the omitted reference period. The coefficients **beta_k** trace out the differential effect of AI exposure at each month relative to the baseline.

Under parallel trends, the pre-period coefficients ($k < 0$) should be statistically indistinguishable from zero. Divergence from zero in the pre-period is direct evidence against the parallel trends assumption. Post-period coefficients ($k > 0$) reveal whether any differential effect persists, grows, or dissipates over time.

4. Core Identifying Assumptions

Assumption 1: Parallel Trends (Conditional) In the absence of ChatGPT, the entry-level share in high-AI-exposure occupations would have trended in parallel with low-exposure occupations, conditional on occupation and time fixed effects. This is the fundamental assumption of the DiD design and is untestable in the post-period. It is partially evaluated in the pre-period via the event-study coefficients.

Assumption 2: No Anticipation Employers did not adjust entry-level hiring in anticipation of ChatGPT's release before October 2022. While OpenAI's GPT-4 capabilities were known to researchers earlier, ChatGPT's November 2022 public launch represented a discrete and widely covered event that plausibly constitutes a sharp break in employer awareness and adoption. This assumption implies that pre-period event-study coefficients should be zero.

5. Threats to Identification and Responses

Threat 1: The Tech-Sector Contraction (Primary Threat) The most serious threat is that the 2022–2023 technology sector downturn — driven by rising interest rates, post-pandemic correction, and overexpansion — coincided precisely with ChatGPT's launch. Because AI-exposed occupations are disproportionately in tech, a decline in entry-level hiring in high-exposure occupations may reflect this cyclical contraction rather than the effect of generative AI.

Response: I will estimate the model on a restricted sample that excludes computer and information technology SOC major groups (SOC 15-xxxx) to test whether the result holds among non-tech AI-exposed occupations (e.g., court reporters, translators, writers, payroll clerks, medical transcriptionists). I will also control for industry-level employment trends interacted with post, if industry-level controls are available.

Threat 2: Pre-Existing Differential Trends The indexed event-study figure (Figure 2 in the current writeup) shows that high-exposure occupations were already declining relative to low-exposure occupations before October 2022, with convergence occurring by mid-2022. This divergence in pre-trends is evidence against the parallel trends assumption for the standard TWFE estimator.

Response: I will augment the TWFE specification with occupation-specific linear time trends to absorb differential pre-existing trends:

$$Y_{it} = \alpha + \beta(\text{Post}_t \times E_i) + \gamma_i + \delta_t + \theta_i t + \epsilon_{it}$$

where θ_i is an occupation-specific linear trend estimated from the pre-period. This allows each occupation to follow its own pre-existing trajectory and identifies β from deviations from that trend after October 2022. I will also consider restricting the estimation window to a narrow band around the event (e.g., 12 months before and after) to limit the influence of longer-run trends.

Threat 3: Measurement Error in AI Exposure The gpt4_beta score is constructed from task-level expert assessments and GPT-4 prompting, not from direct measures of AI adoption. Occupations may have been misclassified, and the score may not accurately reflect which occupations were most rapidly adopting generative AI tools post-2022. Classical measurement error would attenuate β toward zero (downward bias), while systematic misclassification could bias estimates in either direction.

Response: I will estimate the model using alternative exposure measures, including the raw keyword-based AI exposure rate constructed from Lightcast skills data (the self-constructed measure in the earlier version of this project), and compare results across measures. Consistency across exposure operationalizations strengthens confidence in the finding.

6. Planned Robustness Checks

6.1 Placebo Treatment Dates I will re-estimate the DiD using false treatment dates (e.g., October 2019, October 2020, October 2021) and confirm that the interaction $\text{Post}_t \times E_i$ is not significant at these placebo dates. A significant placebo effect would suggest the main result is driven by pre-existing differences rather than the ChatGPT shock.

6.2 Excluding Tech Occupations To address the tech-sector contraction threat, I will re-estimate the model after dropping all SOC 15-xxxx (Computer and Mathematical) occupations. If the estimated beta is similar in magnitude and significance among non-tech AI-exposed occupations (e.g., writers, clerks, transcriptionists), the result is less likely to be driven by tech-sector-specific factors.

6.3 Alternative Exposure Measures I will substitute the `gpt4_beta` score with: (i) the self-constructed keyword-based AI exposure rate; (ii) the `gpt4_alpha` score from Brynjolfsson et al. (2025) as a sensitivity check; and (iii) a binary high/low classification based on the top and bottom terciles of `gpt4_beta`, to confirm that results are not sensitive to the choice or scaling of